

Taicheng Guo

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Email: tguo2@nd.edu

Homepage

Google Scholar

Github

RESEARCH INTERESTS

My long-term research goal is to build fundamental and novel systems for AGI. My research centers on **Agents** and **Systems**, and how they co-evolve on the path to superintelligence, especially Long-Horizon Agent Training, Automated Recursive Self-Improvement/Systems (Code Intelligence, AutoResearch), Agent-System Co-Design, and Language Model Post-Training. I work as a research engineer who is driven to build novel & fundamental systems from scratch.

EDUCATION

University of Notre Dame

US

Ph.D. in Computer Science and Engineering

Advisor: Prof. Xiangliang Zhang. GPA: 4.0/4.0

Thesis: Equipping and Incentivizing Language Models for Research-Level Reasoning Tasks

Committee: Drs. Nitesh V. Chawla, Meng Jiang, Wei Wang, Chandan K. Reddy.

King Abdullah University of Science and Technology (KAUST)

Saudi Arabia

M.S. in Computer Science

Advisor: Prof. Xiangliang Zhang. GPA: 3.89/4.0

Xidian University

Bachelor in Software Engineering. GPA: 3.5/4.0

WORK&INTERN EXPERIENCE

Amazon (Applied Scientist Intern)

Jan. 2025 – Jul. 2025

Worked with Drs. Hai Wang, Xin Chen, Chandan K. Reddy

Topic: Training Long-Horizon Coding Agents.

[\[ACL 2026 Main\]](#), [\[Received Full-Time Return Incline Offer\]](#)

Amazon (Applied Scientist Intern)

Jun. 2024 – Sep. 2024

Worked with Drs. Chaochun Liu, Xin Chen, Chandan K. Reddy

Topic: Building Long-Horizon Agents within Database for Personalization. [\[In Submission\]](#)

Mohamed bin Zayed University of Artificial Intelligence: MBZUAI (Research Intern)

Worked with Drs. Shangsong Liang

Aug. 2022 – Oct. 2022

Topic: Conditional-Based Diffusion Model. [\[ICASSP 2024\]](#)

KAUST (Visiting Student)

Dec. 2020 – Aug. 2021

Worked with Prof. Xiangliang Zhang

Topic: Training Language Models for Cross-Lingual News Recommendation. [\[WWW 2023 Oral\]](#)

Tuyoo Games, AI Platform (Senior Research Engineer)

Aug. 2018 – Oct. 2020

Topic: Led a 4-person team on game-AI intelligence, building zero-sum poker game agents via Monte Carlo Tree Search (MCTS), Minimax, etc.

[\[Employee of the Year Award \(Top 2%\), Employee of the Quarter Award twice \(Top 2%\)\]](#)

Qunar Topic: Strategy Product Manager (Build Personalization Product)

Jul. 2017 – May. 2018

ACADEMIC COMPETITIONS

KDD Cup Large-Scale RecSys Competition. (2nd / 1895 teams, \$4,000) [\[Link\]](#)

NeurIPS Black-Box AutoML Optimization Challenge. (2nd / 200+ teams, W.S. track) [\[Link\]](#)

Kaggle Text Sentiment Analysis. (5th / 80+ teams) [\[Link\]](#)

IEEE Computer Society Data Science challenge. (1st place, \$1000) [\[Link\]](#)

SELECTED PUBLICATIONS

▸ Citations: 2878 (80%+ from first-author papers)

Full publication can be found at: [link](#)

[1] [\[Long-Horizon Agent Training, Automated Recursive Self-Improvement / Systems\]](#)
AutoLLMResearch: Training Research Agents for Automating LLM Experiment Configuration – Learning from Cheap, Optimizing Expensive

Taicheng Guo, Nitesh V. Chawla, Olaf Wiest, Xiangliang Zhang

In Submission

[2] [\[Agent-System Co-Design\]](#) **ASAP: Agent-System Co-Design for WallTime-Centered Auto HPO Research**

Taicheng Guo, Haomin Zhuang, Kehan Guo, Yujun Zhou, Nitesh V. Chawla, Olaf Wiest, Xiangliang Zhang

In Submission

[3] [\[Long-Horizon Agent Training, Automated Recursive Self-Improvement / Systems\]](#) **MTSQL-R1: Towards Long-Horizon Multi-Turn Text-to-SQL via Agentic Training**

Taicheng Guo, Hai Wang, ChaoChun Liu, Mohsen Golalikhani, Xin Chen, Xiangliang Zhang, Chandan K. Reddy

In ACL 2026 Main, Acceptance rate: 19%, 40+ Github Stars

[4] [\[Agent-System Co-Design\]](#) **Large Language Model based Multi-Agents: A Survey of Progress and Challenges**

Taicheng Guo, Xiuying Chen, Yaqi Wang, Ruidi Chang, Shichao Pei, Nitesh V. Chawla, Olaf Wiest, Xiangliang Zhang

In IJCAI 2024, Acceptance rate: 20%, Top-1 Most Influential Paper, 1k+ Github Stars

[5] [\[Language Model Post-Training\]](#) **ReactionTeam: Teaming Experts for Divergent Thinking Beyond Typical Reaction Patterns.**

Taicheng Guo, Changsheng Ma, Xiuying Chen, Bozhao Nan, Kehan Guo, Shichao Pei, Nitesh V. Chawla, Olaf Wiest, Xiangliang Zhang

In IEEE Conference on BigData 2025 Oral, Acceptance rate: 18.7%

[6] [\[LLM Reasoning\]](#) **What can Large Language Models do in chemistry? A comprehensive benchmark on eight tasks**

Taicheng Guo, Kehan Guo, Bozhao Nan, Zhenwen Liang, Zhichun Guo, Nitesh V. Chawla, Olaf Wiest, Xiangliang Zhang

In NeurIPS 2023, Acceptance rate: 32.7%, Top 2% Influential Paper, 150+ Github Stars

[7] [\[Language Model Post-Training\]](#) **Few-shot News Recommendation via Cross-lingual Transfer**

Taicheng Guo, Lu Yu, Basem Shihada, Xiangliang Zhang

In WWW 2023 Oral, Acceptance rate: 19.2%

[8] [\[LLM RL Training, Reward Hacking\]](#) **Domain-Adaptive Reasoning model for Injury Classification from Incident Descriptions**

Taicheng Guo, Hai Wang, Shucheng (Sue) Zhuang, Mohsen Golalikhani, Chaochun Liu, Wei Wang, Xin Chen

In AMLC 2025 RL workshop, Cross-team collaboration work

[9] [\[Agent-System Co-Design\]](#) **Knowledge Graph Enhanced Language Agents for Recommendation**

Taicheng Guo, Chaochun Liu, Hai Wang, Varun Mannam, Fang Wang, Xin Chen, Xiangliang Zhang, Chandan K. Reddy

In Submission

[10] [\[Automated Recursive Self-Improvement / Systems\]](#) **Data Interpreter: An LLM Agent For Data Science**

Sirui Hong, ... Taicheng Guo as a contributor to Graph-based Agent Planning.

In ACL Findings 2025, 68k+ Github Stars